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SECTION A

Answer **all** *questions.*

1. *(a)* Describe an experiment that uses various absorbers to identify the type(s) of radiations emitted by a radioactive sample. Justify all steps in the experiment. [4]

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(b) A radioactive sample of material has a half-life of 11.4 days and an initial activity of A_0. Determine:

(i) the decay constant; [2]

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(ii) the activity of the sample after 57.0 days in terms of A_0; [2]

(iii) the **percentage decrease** in the number of nuclei in the sample after 57.0 days. [3]

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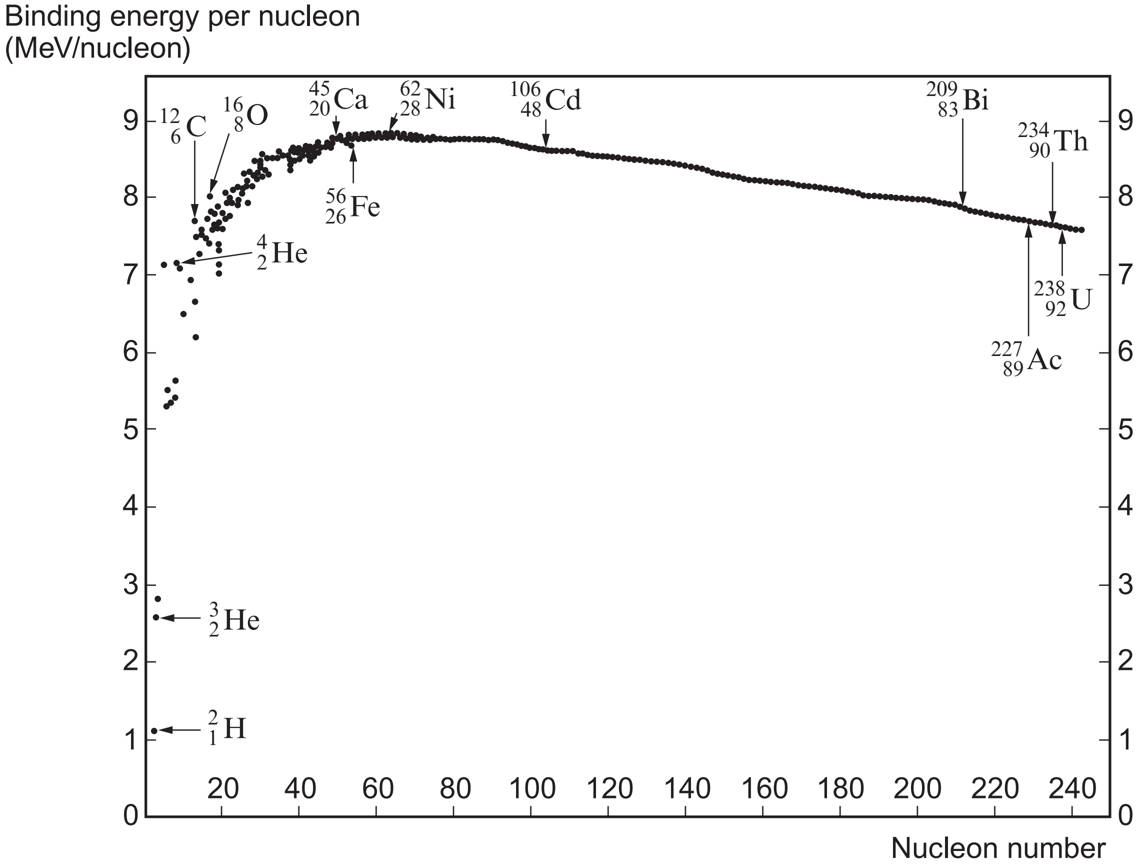
2. *(a)* Nuclear energy can be released by fusion or fission. Describe these processes by considering the graph of the binding energy per nucleon plotted against the nucleon number. [4]



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(b) The fusion of hydrogen to helium in the Sun may be represented by:

$$4\,{}^{1}_{1}\mathrm{H} + 2\mathrm{e}^{-} \rightarrow {}^{4}_{2}\mathrm{He} + 2\,\nu_{e}$$

Particle	Mass / u
${}^{1}_{1}\mathrm{H}$	1.00728
${}^{4}_{2}\mathrm{He}$	4.00151
e^{-}	0.00055
ν_{e}	0.00000

1 u = 931 MeV

Calculate the energy released in this reaction in MeV. [3]

(c) Nuclear power has played a role in the generation of electricity in Wales. Wylfa Newydd on Anglesey may feature in further developments. Indicate a benefit and an issue that may arise from such projects and discuss their relative importance. [3]

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4. *(a)* A fission nuclear reaction for uranium is:

$$^{235}_{92}\mathrm{U} + ^{1}_{0}\mathrm{n} \longrightarrow ^{92}_{36}\mathrm{Kr} + ^{141}_{56}\mathrm{Ba} + 3^{1}_{0}\mathrm{n}$$

The masses of the nuclei are:

mass of $^{235}_{92}\mathrm{U}$ = 235.01 u

mass of $^{92}_{36}\mathrm{Kr}$ = 91.90 u

mass of $^{141}_{56}\mathrm{Ba}$ = 140.89 u

mass of $^{1}_{0}\mathrm{n}$ = 1.01 u

1 u = 931 MeV

Calculate the energy released in this nuclear reaction. [2]

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(b) Explain carefully the relevance of the **binding energy per nucleon** curve to nuclear fission and fusion. [6 QER]

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5. *(a)* Radon gas decays by emitting α-particles. It has a half-life of 3.8 days. Calculate the **percentage reduction** in the activity of a sample of radon after 12 days. [4]

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(b) A student makes the following measurements for a radioactive source using the indicated absorber between the source and detector.

Absorber	Counts per minute
none	1 004
sheet of paper	597
2 mm of aluminium	23
15 cm of lead	27

Explain these observations. [4]

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6. *(a)* Complete the following decay equations.

(i) $^{228}_{90}\text{Th} \longrightarrow \,^{\cdots\cdots}_{\cdots\cdots}\text{Ra} + \,^{\cdots\cdots}_{\cdots\cdots}\alpha$ [1]

(ii) $^{90}_{38}\text{Sr} \longrightarrow \,^{\cdots\cdots}_{\cdots\cdots}\text{Y} + \,^{\cdots\cdots}_{\cdots\cdots}\beta$ [1]

(b) Calculate the binding energy **per nucleon** for $^{90}_{38}\text{Sr}$. [4]

m_{proton} = 1.007 276 u

m_{neutron} = 1.008 664 u

m_{electron} = 0.000 549 u

atomic mass of $^{90}_{38}\text{Sr}$ = 89.907 738 u

1 u = 931 MeV

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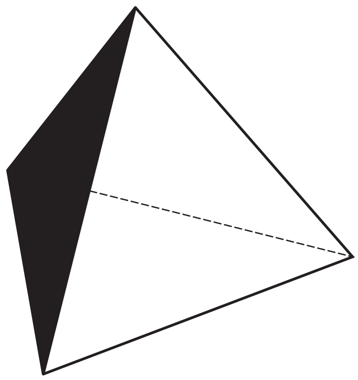
(c) Radioactive decay may be investigated in the laboratory using a dice analogy.

A student has 564 identical wooden dice. These dice are tetrahedrons, with one face painted black. A tetrahedron is a solid with four identical faces, each face being an equilateral triangle.



(i) If the student throws one of the dice, what is the probability of the tetrahedron landing on the black face? [1]

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(ii) The student then throws all dice and counts and discards those that landed on the black face. The remaining dice are thrown again and the process is repeated multiple times. The number of dice discarded after each throw is recorded in the table, and the number remaining after each throw is calculated. The results are shown in the table, which also gives the fraction of dice remaining, where:

$$\text{Fraction of dice remaining after a throw} = \frac{\text{number of dice remaining after a throw, } R}{\text{number of dice thrown in that throw, } T}$$

Throw number, n	Number of dice thrown, T	Number of dice discarded after throw	Number of dice remaining after throw, R	Fraction remaining after throw, $\frac{R}{T}$
1	564	138	426	0.76
2	426	116	310	0.73
3	310	87	223	0.72
4	223	52	171	0.77
5	171	39	132	0.77
6	132	34	98	0.74
7	98	10	88	0.90
8	88	27	61	0.69
9	61	18	43	0.70
10	43	8	35	0.81

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I. Explain why the expected fraction remaining after n throws is $(0.75)^n$. [2]

II. Hence predict the expected number remaining after 10 throws. [1]

III. Discuss to what extent the observations agree with those expected from theory. [3]

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SECTION A

Answer **all** questions.

1. (a) Radon is a radioactive gas and emits alpha particles.

(i) Explain what is meant by the *activity* of a radioactive source **and** give its unit. [2]

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(ii) The half-life of radon is 3.3×10^5 s. Determine the time it takes for a sample of radon to decay to 20 % of its initial activity. [3]

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(iii) Discuss the extent to which alpha particles are a risk to the human body. [3]

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(b) Radium-226 decays into radon.

$$^{226}_{88}\text{Ra} \longrightarrow {}^{222}_{86}\text{Rn} + {}^{4}_{2}\alpha$$

Assuming that 98% of the energy released in this process is converted to the kinetic energy of the alpha particle, use the data below to determine the speed of the alpha particle. [5]

Atomic masses: $^{226}_{88}\text{Ra}$ = 226.025410 u $^{222}_{86}\text{Rn}$ = 222.017578 u

$^{4}_{2}\text{He}$ = 4.002603 u m_{electron} = 0.000549 u

1 u = 931 MeV

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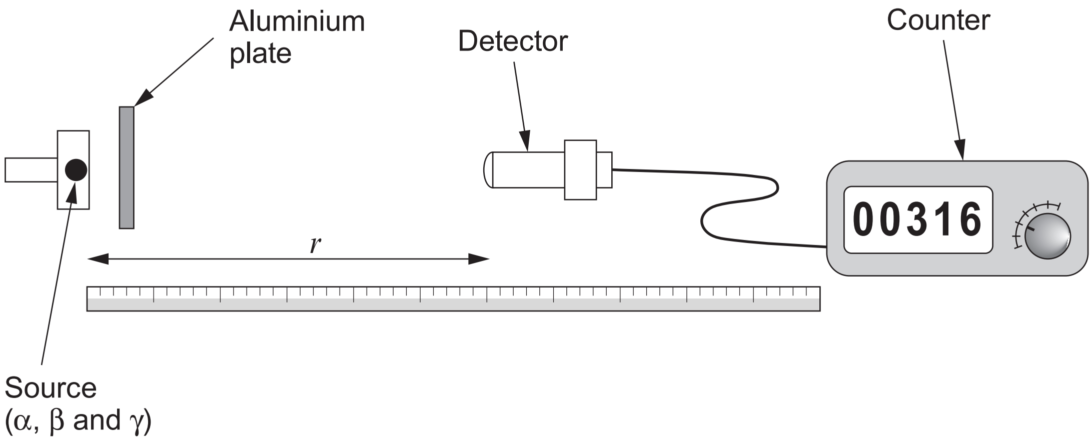
5. A student says that the radiation intensity, I, of gamma rays increases with decreasing distance, r, from a source such that:

$$I \propto \frac{1}{r^n}$$

She wishes to determine the value of n.

The student uses a source of alpha, beta and gamma radiation. She measures the intensity of the rays at set distances from the source by determining the count over intervals of one minute. She also takes measurements without the source at the start and at the end of the experiment.



(a) (i) Suggest why a thin aluminium plate is placed near the source. [1]

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(ii) Explain why readings are taken without the source, at the start **and** at the end of the experiment. [2]

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(b) At each distance, r, the counts were measured over one minute intervals three times. The uncertainty in r is ± 0.005 m. The measurements are recorded in the table.

Complete the table. [2]
Space for calculations.

Distance, r / m	Count				Uncertainty in mean count
	First reading	Second reading	Third reading	Mean	
without source	19	22	25	22	3
1.000	110	130	136	125	13
0.800	195	165	178	....	....
0.600	270	316	300	....	....
0.400	661	604	651	639	29
without source	20	20	25	22	3

(c) (i) The student subtracts the mean count without the source from the mean count for each distance to obtain the corrected mean count, N. She also determines the uncertainty in each of these corrected mean counts.

Complete the table. [2]
Space for calculations.

Distance, r / m	Corrected mean count, N	Uncertainty in corrected mean count
1.000	103	16
0.800	....	....
0.600	....	....
0.400	....	....

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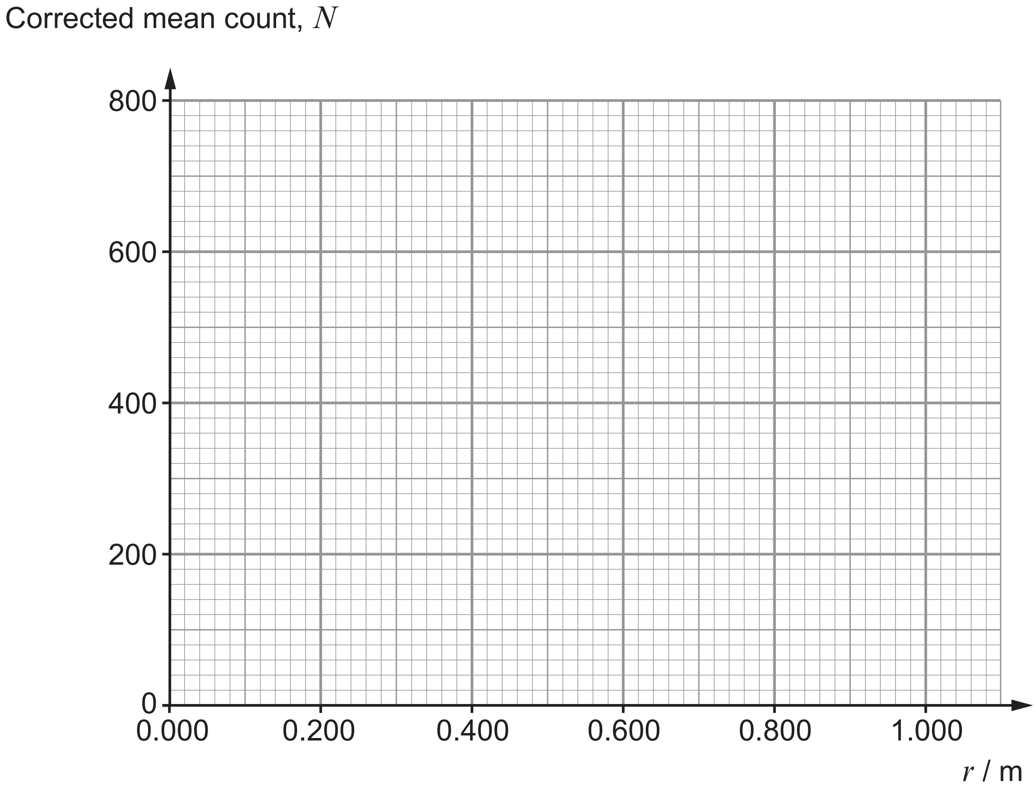
(ii) Plot a graph of the corrected mean count, N, against distance, r. **Error bars are not required.** [3]



(iii) The student claims that error bars may be shown on the graph for the corrected mean count, but not the distance. Justify her claim. Calculations are not required. [1]

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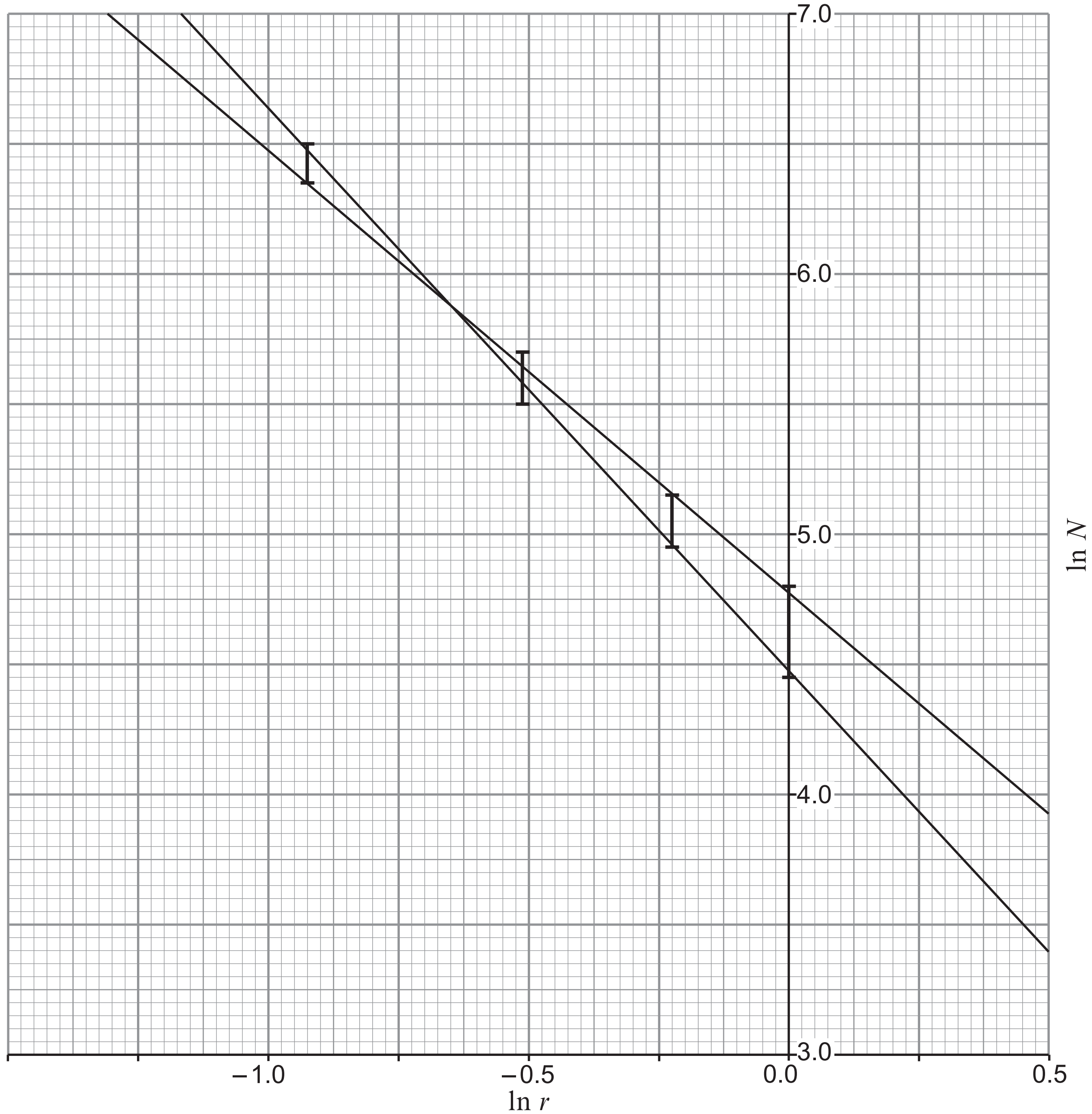
(d) Her friend suggests using a logarithmic graph. She calculates the logarithmic values in the table below.

Distance, r / m	$\ln r$	$\ln N$
1.000	0.00	4.63
0.800	–0.22	5.06
0.600	–0.51	5.61
0.400	–0.92	6.42

The data are plotted on the graph.



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For a distance of r = 1.000 m use the table in part (c) to show that the uncertainty in the logarithmic counts for this distance is approximately 0.15. [3]

(e) (i) Starting with intensity: $I \propto \frac{1}{r^n}$ justify how the logarithmic graph can be used to determine n. [3]

(ii) Use the maximum and minimum gradients to determine n and the **absolute** uncertainty in its value. [4]

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3. (a) State what is meant by alpha emission. [2]

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(b) Americium-241 has a half-life of 432 years. Determine the percentage **decrease** in the activity of the element after 30 years. [4]

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(c) A builder is asked to install a smoke detector. He researches smoke detectors and finds that the radioactive source in the detector emits alpha particles. He decides that it is too dangerous to install the detector. Discuss whether or not he has used science well in his decision making. [2]

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(d) Describe how alpha, beta and gamma radiation can be distinguished:

- by using different absorbers
- by using an electric field **or** a magnetic field. [6 QER]

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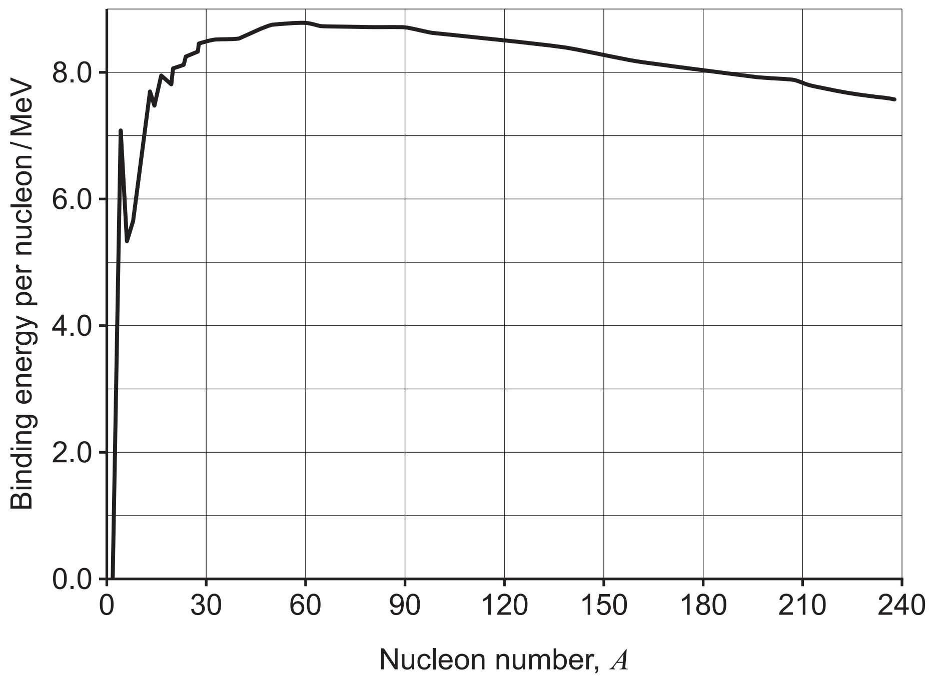
5. (a) Nuclear energy can be released by fusion and fission. Discuss these processes in terms of energy and the stability of the nuclei. Use the graph of the binding energy per nucleon in your answer. [4]



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(b) One of the many possible fission reactions of uranium-235 is:

$$^{235}_{92}\mathrm{U} + ^{1}_{0}\mathrm{n} \rightarrow ^{98}_{40}\mathrm{Zr} + ^{135}_{52}\mathrm{Te} + 3\,^{1}_{0}\mathrm{n}$$

The masses are:

$^{235}_{92}\mathrm{U}$: 235.04393 u; $^{98}_{40}\mathrm{Zr}$: 97.91273 u; $^{135}_{52}\mathrm{Te}$: 134.91645 u, $^{1}_{0}\mathrm{n}$: 1.00866 u
and 1 u = 931 MeV.

(i) State what is meant by the Avogadro constant. [1]

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(ii) Calculate the energy released by the fission of 0.0010 kg of uranium-235 in this reaction. Give your answer in J. [5]

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SECTION A

Answer **all** questions.

1. A fission reaction of uranium is:

$$ {}^{1}_{0}\mathrm{n} + {}^{235}_{92}\mathrm{U} \longrightarrow {}^{141}_{56}\mathrm{Ba} + {}^{92}_{36}\mathrm{Kr} + 3\,{}^{1}_{0}\mathrm{n} $$

Masses:

${}^{235}_{92}\mathrm{U}$ = 235.043 930 u ${}^{141}_{56}\mathrm{Ba}$ = 140.914 412 u ${}^{92}_{36}\mathrm{Kr}$ = 91.926 156 u

m_{neutron} = 1.008 665 u 1 u = 931 MeV

(a) Calculate the energy released in this reaction giving your answer in MeV. [4]

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(b) If 1.2 kg of ${}^{235}_{92}\mathrm{U}$ undergoes this reaction, show that the total energy released is approximately 8.5×10^{13} J. [3]

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(c) An engineer at a nuclear power station claims that over two billion 11 W light bulbs could be powered for 1 hour by the energy released in part (b). Discuss whether or not her claim is correct. [One billion = 1×10^9] [3]

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3. (a) Describe an experiment to investigate the variation of intensity of gamma radiation with the distance from a source. Consider the experimental set-up, data collection and analysis to confirm an inverse square law relationship (space has been provided for a diagram). [6 QER]

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(b) Gamma radiation is known to cause cancer and microwaves are thought by some to be linked to brain cancer.

(i) Calculate the energy of a photon of:

I. gamma radiation of wavelength 1.0 pm. [2]

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II. microwave radiation of frequency 1.8 GHz. [1]

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(ii) Some people claim that mobile phone usage increases the rate of brain cancer, while others say that it does not cause significant damage. What steps can be taken to investigate who is correct? [2]

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6. The beta decay of ${}^{14}_{6}C$ can be used for dating organic samples. The half-life of ${}^{14}_{6}C$ is 5730 years.
The percentage of ${}^{14}_{6}C$ atoms in a piece of wood from an old boat is found to have **decreased by** 30% since it was formed from living material.

(a) Estimate the age of the boat. [4]

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(b) Calculate the activity of the ${}^{14}_{6}C$ after 7000 years as a percentage of its original activity. [3]

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